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Exercise and the Regulation of Skeletal Muscle Hypertrophy

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Abstract

Skeletal muscle is a critical organ serving as the primary site for postprandial glucose disposal and the generation of contractile force. The size of human skeletal muscle mass is dependent upon the temporal relationship between changes in muscle protein synthesis (MPS) and muscle protein breakdown. The aim of this chapter is to review our current understanding of how resistance exercise influences protein turnover with a specific emphasis on the molecular factors regulating MPS. We also will discuss recent data relating to the prescription of resistance exercise to maximize skeletal muscle hypertrophy. Finally, we evaluate the impact of age and periods of disuse on the loss of muscle mass and the controversy surround the etiology of muscle disuse atrophy.



1. INTRODUCTION

Exercise *per se* is a potent stimulus for the remodeling and reconditioning of skeletal muscle.¹ Alterations in the load, frequency, and duration of contractile activity are known to induce differential phenotypic adaptations specific to the nature of the stimulus.¹ Resistance exercise, for

example, is characterized by repeated bouts of loaded contractions that result in the rapid recruitment of type 2 muscle fibers and stimulation of muscle protein synthesis (MPS).^{1,2} When combined with adequate nutrition, resistance exercise improves muscular strength partly through an increase in the radial diameter of individual muscle fibers, a process termed *hypertrophy*.³ At the outset, it is important to first clarify the nomenclature of the term *hypertrophy*. The increase in skeletal muscle size with resistance exercise is due to an increase in the cross-sectional area (CSA) of individual muscle fibers. There are, however, documented reports—mainly in rodents—of increases in muscle size associated with increases in the number of individual myofibers within a muscle,⁴ which is termed *hyperplasia*. Hyperplasia is proposed to be the result of satellite cell proliferation and new fiber formation, but to date, there is little empirical evidence that hyperplasia occurs in physiological models of human resistance exercise. That is not to say that satellite cells are not important in exercise adaptation. The primary aim of this chapter is to review our current understanding of the mechanisms by which resistance exercise stimulates skeletal muscle *hypertrophy*. We will begin by discussing the biological basis of resistance exercise-induced muscle *hypertrophy* and a degree of incongruence between current recommendations and recent data relating to prescription of resistance exercise training. We will then examine the variability in the hypertrophic response to resistance training between individuals and how, at the molecular level, this variability and MPS in general are regulated. Specifically, we will discuss the influence of factors intrinsic to skeletal muscle such as the activation of the mechanistic target of rapamycin complex 1 (mTORC1) upon skeletal muscle loading. We will conclude by describing how muscle unloading influences the regulation of muscle mass. For the purposes of concision and relevance to the human model, we will focus on data procured primarily from humans; however, when necessary, studies from rodent and cell-based systems will be incorporated to support relevant points of discussion.



2. RESISTANCE EXERCISE TO ENHANCE SKELETAL MUSCLE MASS

Human skeletal muscle mass is a function of the balance between MPS and muscle protein breakdown (MPB). In healthy nondiseased humans, it is the changes in the rates of MPS and not those of MPB that influence myofiber size. It is for this reason that the focus of the majority of research in recent years has been on factors that ultimately impact upon MPS. Many

of these studies have employed stable isotope technology to directly track the incorporation of labeled amino acids into new muscle proteins and into the contractile apparatus of a myofiber. Since the introduction of this technology into the exercise and nutritional sciences, a large body of data has been generated characterizing how various exercise and nutritional manipulations influence muscle protein turnover. In this regard, we now know that the consumption of proteins or amino acids, particularly the essential amino acids, leads to a systemic hyperaminoacidemia stimulating a significant and transient increase in rates of MPS.⁵ When a bout of resistance exercise is performed prior to the consumption of amino acids, there is a synergistic impact on rates of MPS over and above that observed with amino acid ingestion alone.⁶ In essence, resistance exercise sensitizes skeletal muscle to the anabolic effects of amino acid consumption, an effect that can persist for up to 24–48 h postexercise.⁷

One of the earliest documented reports that resistance exercise enhances muscle function was by De Lorme and Watkins who, in 1948,⁸ prescribed “progressive resistance exercise” to treat patients with poliomyelitis. Since this work and that of others demonstrating the efficacy of resistance exercise to enhance skeletal muscle mass,⁹ resistance exercise training is now widely used in both clinical and athletic settings to combat diseases associated with the loss of muscle mass (myopenia) and strength (dynapenia) as well as to enhance athletic success and general fitness. The term progression refers to the appropriate manipulation of (i) load per repetition, which is most often prescribed as a percentage of one repetition maximum (1 RM); (ii) number of sets performed; (iii) recovery between sets; (iv) muscle action (concentric vs. eccentric); and (v) total exercise volume (i.e., load $\times$ repetitions). Despite receiving significant experimental attention, the optimal resistance exercise–training program incorporating various modifications of the above variables, at least from the perspective of inducing skeletal muscle hypertrophy, remains unknown, and it could be argued that on a population basis it may not even exist.

Current recommendations from the most recent American College of Sport Medicine position stand¹⁰ state that for “...novice and intermediate individuals, it is recommended that moderate loading be used (70–85% 1 RM) for 8–12 repetitions per set for one to three sets per exercise’. Whilst for advanced trainers ‘it is recommended that a loading range of 70–100% of 1 RM be used for 1–12 repetitions per set for three to six sets per exercise in a periodized manner such that the majority of training is devoted to 6–12 RM and less training devoted to 1–6 RM.” These recommendations

and the literature supporting them are surprisingly thin. In a series of studies, it has been demonstrated that repetition load plays a minimal, if not negligible, role in the hypertrophic response to loaded contractions as long as volitional fatigue is achieved.¹¹ Indeed, in one study recreationally trained participants who performed 30% of their 1 RM until failure, elicited similar increases in MPS 4 h postresistance exercise relative to participants who performed resistance exercise at 90% of their 1 RM. Another interesting finding was that myofibrillar MPS was elevated 24 h postresistance exercise in the 30% 1 RM condition but not the 90% 1 RM condition. Expanding upon these findings, a follow-up study determined whether the similar acute responses (i.e., MPS) translated into longitudinal changes in muscle size.¹² Again, contrary to the existing guidelines, performing resistance exercise to failure with a low repetition load (in this instance 30% 1 RM) produced the same gains in muscle size as compared with performing resistance exercise to failure at a load of 80% of 1 RM. It is important to note that strength gains in that study were greater in the 80% 1 RM group. Thus, from a strength perspective, training at a higher repetition load may be beneficial, particularly for athletes. It also must be acknowledged that exercise regimens characterized by higher exercise volumes (i.e., sets $\times$ repetitions $\times$ sessions) display favorable gains in muscle growth.¹³ So, taken together, these data demonstrate that as long as volitional failure is achieved, and the exerciser performs a comparable amount of effort, repetition load has little influence in the hypertrophic response to resistance exercise at least within the range of 30–80% of 1 RM.¹¹



3. HETEROGENEITY IN RESPONSE TO RESISTANCE TRAINING

As previously discussed, an acute bout of resistance exercise coupled with amino acid ingestion results in an increase in the rate of MPS, whereas frequent, episodic bouts of resistance exercise, and amino acid ingestion promote skeletal muscle hypertrophy.⁹ The increases in muscle growth with resistance training are often cited as occurring at a linear rate; however, we propose that expressing changes in rates of hypertrophy, as an average rate per day likely obscures some important insight into the process of muscular hypertrophy. In fact, there are numerous reports of large variability in both MPS and gains in muscle size in response to resistance training between individuals.^{14,15} The individual variance in gains in muscle mass with resistance training was well demonstrated by Hubal *et al.*¹⁴ in

585 participants with assessment of changes in isometric and dynamic strength as well as changes in muscle CSA following 12 weeks of progressive resistance training of the *biceps brachii*. Intriguingly, in response to the training protocol, 1 RM strength gains ranged from 0% to +250%, whereas changes in muscle size ranged from -2% to +59%. When examining the differences between the sexes, changes in muscle CSA were 2.5% greater in men compared with women, while relative gains in strength were greater in women. There are other studies that also support the notion of extreme individual variability in the hypertrophic response to supervised resistance training (Fig. 1). Indeed, Mitchell *et al.*¹² observed that following 16 weeks of resistance exercise training increases in fat- and bone-free mass ranged from 0 to 9 kg while there was also large variability in the gains in quadriceps volume (443 cm³) and leg press 1 RM (200 kg). These findings are in agreement with those of Hubal *et al.*,¹⁴ suggesting that a “one size fits all” approach is likely neither appropriate nor effective for promotion of optimal gains with resistance training. Another interesting finding from the study of Mitchell *et al.*¹² was that, in addition to the variable hypertrophic response, the increase in MPS following the first bout of resistance exercise failed to predict the magnitude of gains in muscle mass. So, it appears that not only are there differences in the hypertrophic response *between* individuals following resistance exercise training but the magnitude of the response *within* individual also lacks uniformity. Factors extraneous to the control of the study (i.e., nutritional intake) could be cited as confounding factors, but the marked range in skeletal muscle mass gains among humans is now thought to be biological in nature. However, biological

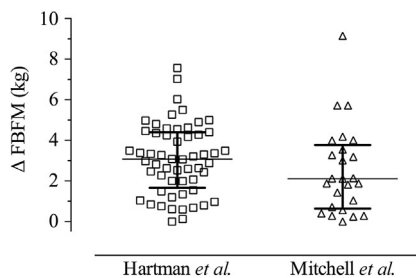


Figure 1 Change in fat- and bone-free mass (FBFM) in response to 10 weeks of resistance exercise training (Mitchell *et al.*¹²) and in response to 12 weeks of resistance training in combination with fat-free milk ($n = 18$), fat-free soy ($n = 19$), and control ($n = 19$) (Hartman *et al.*¹⁶). Data are expressed as individual delta change (kg), median, and interquartile range.

basis for resistance exercise-induced skeletal muscle hypertrophy remains unknown and is currently a topic of intense debate.¹⁷



4. THE INFLUENCE OF SYSTEMIC HORMONES

For many years, it has been known that resistance exercise results in a transient (15–30 min) increase in what are purported to be “anabolic” hormones such as free and protein-bound forms of testosterone, insulin-like growth factor 1 (IGF1), and growth hormone.¹⁸ Although there is evidence to show that exogenous administration of supraphysiological doses of testosterone enhances skeletal muscle hypertrophy,¹⁹ there is comparably little support for the efficacy of growth hormone²⁰ and IGF1²¹ to exert the same effects, at least in adult humans. Indeed, it is indisputable that secretion of growth hormone supports bone and muscle growth in children, and supraphysiological doses of growth hormone enhance collagen protein synthesis.²¹ However, the administration of growth hormone directly does not enhance rates of skeletal MPS.²⁰ One argument is that growth hormone-enhanced collagen synthesis may facilitate increased loading of the muscle during resistance exercise, which, subsequently, stimulates greater rates of MPS. However, this hypothesis has yet to be experimentally corroborated. Alternatively, growth hormone is known to promote IGF1 production in the liver. IGF1 also is elevated in a transient manner postexercise²² and thus has been implicated, in addition to both growth hormone and testosterone, as a mediator of MPS. Data to support this assertion have come predominantly from studies in rodent models. In one such study, blockade of the IGF1 receptor resulted in a reduction in the elevation of contraction-induced rates of MPS,²³ whereas a contrary report demonstrated that deletion of the IGF1 receptor in mice did not influence load-induced skeletal muscle hypertrophy.²⁴ Furthermore, neither IGF1 receptor phosphorylation nor any other marker of receptor tyrosine kinase activation is increased following resistance exercise in rodents,²⁵ suggesting that paracrine-related growth factors are not released in sufficient quantities to impact receptor function. Notwithstanding, the fundamental differences between rodents and humans,^{26,27} it is also important to point out that whereas insulin can induce phosphorylation of the IGF1 receptor, resistance exercise does not.²⁵ Moreover, there is now significant data in human models to show that physiological increases in insulin concentrations, corresponding to those seen with ingestion of a meal, play a permissive, and not a stimulatory role in the regulation of MPS.^{28,29} In our view, the physiological increases in

growth hormone and insulin-like growth factor in response to resistance exercise may be similarly permissive, if they have a role at all, in mediating changes in rates of MPS.

Whereas there is little evidence to support the role of growth hormone in skeletal muscle accretion, the pharmacological administration of testosterone is known to have a direct effect on the accrual of skeletal muscle.¹⁹ Indeed, the suppression of circulating testosterone concentrations—induced by treatment with goserelin—to 10% of circulating concentrations compromises strength gains following 12 weeks of strength training.³⁰ On the contrary, administrations of 600 mg of testosterone enanthate to healthy young humans increased fat-free mass, muscle size, and muscle strength even in the absence of resistance exercise.¹⁹ Importantly, there are profound and physiologically relevant differences between the pharmacological suppression and pharmacological doses of testosterone and the natural increases observed during the postresistance exercise period.³¹ The acute increase in testosterone following a bout of resistance exercise is often ~0.1% to 1% of the dose administered exogenously and acts for comparably shorter durations, depending upon the dosing strategy. A good example of this point is the fact that despite a ~45-fold lower exercise-induced increase in testosterone, women exhibit comparable rises in rates of MPS and hypertrophy as compared to men.³² In addition, the postexercise rise in testosterone does not exceed that of what is known to occur with diurnal variation.³¹

In a study from West *et al.*,³³ the notion that resistance exercise-induced increases in purportedly anabolic hormones play a significant role in skeletal muscle hypertrophy was examined. In this experiment, 12 healthy males were required to perform a series of unilateral arm curls on separate days for 15 weeks. One arm was exercised alone (“low hormone” condition), whereas participants performed a bout of heavy lower-limb resistance exercise, designed to elicit an increase in circulating “anabolic” hormones, on the days that the other arm performed resistance exercise (“high hormone” condition). Interestingly, although there was a significant and profound increase in testosterone, IGF1, and growth hormone in the “high hormone” condition, there was no significant difference in muscle CSA or strength compared to the “low hormone” condition. If it were true that the acute increase in such hormones were playing a role in adaptations to resistance exercise, then the “high hormone” condition would be expected to produce greater gains in muscle CSA and strength, yet they failed to do so. The interested reader is directed to a debate regarding the role that hormonal changes play in resistance exercise-induced hypertrophy.³¹

5. THE ROLE OF THE mTORC1

A central protein complex in signal transduction, mTORC1, is an evolutionarily conserved threonine/serine protein kinase that is critical for cell survival, cell proliferation, and the synthesis of new muscle proteins (Fig. 2). It is composed of a catalytic subunit, mechanistic target of rapamycin (mTOR) as well as adaptor proteins: regulatory-associated protein of mTOR (RAPTOR), mammalian lethal with SEC13 protein 8 (mLST8), DEP domain-containing mTOR-interacting protein (DEPTOR), and proline-rich Akt substrate of 40 kDa (PRAS40).³⁴ There is an array of molecular input signals to mTORC1 that can regulate its activity. One such input is the guanosine triphosphatase (GTPase) Ras homolog enriched in brain (RHEB). When guanosine triphosphate (GTP) is bound to RHEB, there is a RHEB-mediated stimulation of mTORC1 assembly

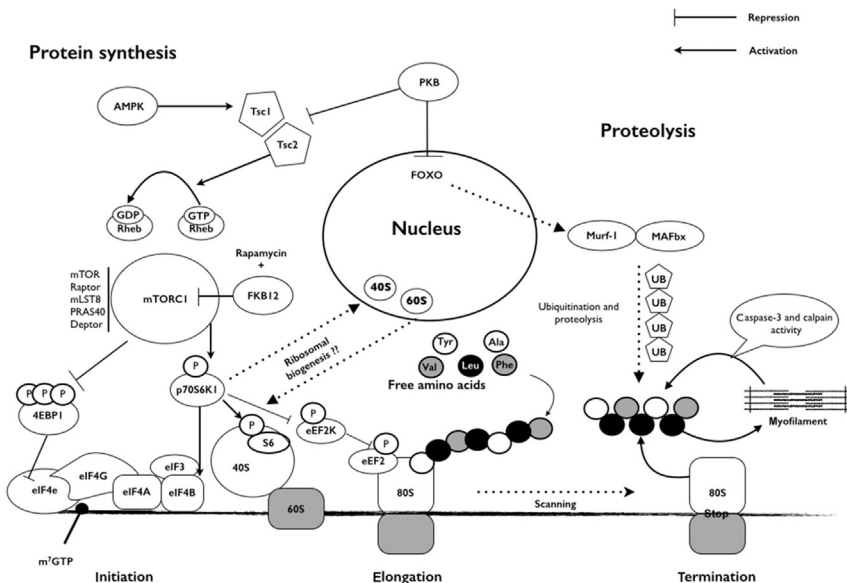


Figure 2 Schematic illustration of the putative molecular factors regulating skeletal muscle protein synthesis and proteolysis. Stimulation of the mechanistic target of rapamycin complex 1 precipitates a series of posttranslational modifications including, but not limited to, the phosphorylation of eukaryotic initiation factors (eIFs), binding proteins, and protein kinases. Upregulation of the ubiquitin proteasome pathway results in the expression of genes encoding for ubiquitin ligases and the subsequent degradation of muscle proteins. Abbreviations and names of genes, mRNAs, and proteins can be found in the text.

and activation.³⁵ This process is further regulated by an upstream GTPase-activating protein (GAP) called tuberous sclerosis complex 2 (TSC2). TSC2 activates RHEB to drive its GTPase activity, increasing the guanosine diphosphate (GDP)/GTP-bound state of RHEB, and thus reducing the ability to activate mTORC1.³⁶ In turn, the influence of TSC2 on RHEB can be directly enhanced by adenosine monophosphate-activated protein kinase (AMPK).³⁷ Moreover, AMPK (an energy stress sensor) also has been shown to inhibit mTORC1 via phosphorylation of DEPTOR.³⁸ Alternatively, protein kinase B (PKB) (an important downstream protein for receptor tyrosine kinases) phosphorylates TSC2 to inhibit its interaction with RHEB³⁹ as well as directly inhibiting the function of the mTORC1 inhibitor PRAS40,⁴⁰ which exerts a stimulatory effect on mTORC1 activity.

Amino acid sufficiency is signaled to mTORC1 via multiple mechanisms including the lipid kinase human vacuolar sorting protein 34, mitogen-activated protein kinase 3, and the RagGTPases.³⁵ In essence, mTORC1 acts like a control center integrating inputs on energy stress, hormonal status, and amino acid sufficiency and relaying that information through its downstream targets to control the rate of MPS. mTORC1 targets many downstream substrates including, but not limited to, 70-kDa ribosomal protein S6 kinase 1 (P70S6K1) and eukaryotic initiation factor 4E-binding protein 1 (4EBP1).⁴¹ Both P70S6K1 and 4EBP1 play a crucial role in orchestrating the MPS response to anabolic stimulation. Indeed, P70S6K1 is phosphorylated by mTORC1 on Thr389 to upregulate the opening round of translation initiation via the S6K1 Ally/Ref-like target (SKAR) protein⁴² and translational elongation via eukaryotic elongation factor-2 kinase (EEF2K).⁴³ The function of 4EBP1 is inhibited by mTORC1 (phosphorylation occurs on Thr37/46), which relieves the inhibition on eukaryotic initiation factor 4E (EIF4E). This phosphorylation event enables EIF4E to interact with eukaryotic initiation factor 4G (EIF4G) key to the commencement of protein initiation.⁴⁴ It is noteworthy that there are other signaling molecules that either regulate or are targets of, mTORC1, and we acknowledge the existence of another mTOR complex, mechanistic target of rapamycin complex 2 (mTORC2). But the role of mTORC2 and the aforementioned unrelated signaling events are outside the purview of the present discussion, so we refer the interested reader to several informative papers.^{45–47}

Given that translation initiation is proposed to be orchestrated by mTORC1-4EBP1/P70S6K1 signaling,⁴⁸ it is unsurprising that increased rates of MPS in response to resistance exercise (RE) are often accompanied

by elevations in the phosphorylation status of these molecules.^{49–53} However, many studies only show associations or correlations between changes in protein phosphorylation (and presumably activation) and rates of MPS, not a direct cause and effect relationship. One of the first investigations to show that mTORC1 was critical for mechanical overload-induced increases in muscle hypertrophy was conducted in rodents by Bodine *et al.*⁵⁴ In this classic study, the authors showed that treatment with the highly specific mTORC1 inhibitor, rapamycin, repressed synergist ablation-induced increases in skeletal muscle hypertrophy as well as PKB-mTORC1-P70S6K1 signaling. Other researchers expanded upon these findings in mice, using various rapamycin resistant mutants of mTOR to show that rapamycin also reduced mTORC1 signaling and individual muscle fiber CSA following chronic mechanical loading.⁵⁵ However, it is important to note that there are significant interspecies differences between rodents and humans with regards to protein turnover²⁶ that should be acknowledged when interpreting these data. For example, the work of Goodman *et al.*⁵⁵ showed that chronic mechanical loading also induced a degree of hyperplasia, a phenomenon that, to our knowledge, does not occur in human models of resistance exercise training. Nevertheless, what these data show, at least in rodent models, is that loading-induced skeletal muscle hypertrophy occurs predominantly in an mTORC1-dependent manner.

To date, only a few studies in humans have used rapamycin to examine if RE-induced increases in MPS are reliant on mTORC1-mediated signaling.^{56,57} By administering rapamycin 2 h prior to a bout of RE, the authors of one study were able to significantly raise blood rapamycin concentrations at the commencement of exercise. Skeletal MPS was determined using the stable isotope technique 1 and 2 h post-RE during which the phosphorylation status of mTOR and associated signaling agents was measured. Rapamycin significantly reduced MPS during the initial 1–2 h post-RE recovery period. Moreover, the reduction in MPS was accompanied by a reduction in mTOR^{ser2448} and P70S6K1^{thr389} phosphorylation at 1 h post-RE. These data show that activation (phosphorylation) of mTOR signaling plays a key role in contraction-induced increases in human skeletal MPS, a finding that corroborates earlier reports in rodents.^{54,55} Despite a reduction in MPS, rapamycin treatment did not inhibit phosphorylation of P70S6K1 2 h post-RE nor did it have any impact on 4EBP1 phosphorylation at any time point during the recovery period. The latter finding is not entirely surprising given that 4EBP1 exhibits resistance to the effects of rapamycin,⁵⁸ and the dose of rapamycin employed was, in comparison to

those as delivered in rodent experiments, relatively low. These data may also suggest that multiple redundant signaling pathways are required to maximize rates of MPS in response to RE. Clearly, more work in human models of exercise is now needed to define the exact molecular mechanisms by which RE stimulates a rise in the rates of human MPS.

A pertinent question is, since mTORC1 is a focal point controlling cell size in response to anabolic stimulation, what signals regulate mTORC1? We now know that mTORC1 activation in response to contraction occurs in a 3-phosphoinositide-dependent protein kinase (PI3K)-PKB-independent manner, and exercise-induced increases in circulating “anabolic” hormones (IGF1 and testosterone) also play little-to-no role. One candidate that has received much attention is focal adhesion kinase (FAK), which is an integrin protein located within the costameres of skeletal muscle and has exhibited mechanical sensitivity in rodent, cell, avian, and human models of loaded contraction.^{59–62} The sensitivity of FAK to mechanical loading was highlighted in a study by Fluck *et al.*,⁶² who identified an increase in FAK activity following chronic overload of avian skeletal muscle. Furthermore, in a study from our laboratory, 14 days of limb unloading via limb immobilization in humans reduced FAK^{Tyr576/577} phosphorylation,⁵⁹ a finding that is consistent with another report in a human immobilization model.⁶³ While informative, these studies do not provide mechanistic insight into how FAK activity, in response to contractile perturbation, regulates MPS. There is evidence that FAK is required for IGF1-mediated increases in protein synthesis in myotubes via TSC2-mTORC1-P70S6K1 signaling.⁶⁴ However, in humans, we know that IGF1 plays a minimal role in RE-induced increases in MPS. Other candidate signaling molecules that have been shown to impact mechanically sensitive mTORC1 activity in rodent models include phospholipase D (PLD)⁶⁵ and diacylglycerolkinase zeta,⁶⁶ but whether these molecules are responsible for load-induced modulation of mTORC1 and MPS remains a topic of debate. Clearly, more work in humans is needed to identify how mechanical load is transmitted to a biochemical signal that is capable of mediating contraction-induced increases in MPS.

In nearly all of studies of signaling pathway activity, particularly the human studies, the traditional method to assess the activity or functional relevance of a protein in response to exercise and or feeding is to measure its posttranslational phosphorylation status by immunoblotting (Western blotting) or (γ -³²P) adenosine triphosphate (ATP)-dependent kinase assays.⁶⁷ Emergent data using immunohistochemistry have also provided information

relating to the subcellular trafficking of mTOR and this approach has yielded exciting information.^{68,69} Indeed, the electrical stimulation of mouse skeletal muscle to elicit eccentric contraction demonstrated that changes in the phosphorylation status of TSC2 within an RxRxx motif resulted in its translocation away from the lysosomal surface.⁶⁸ Given that the mTORC1 regulator RHEB is located at the lysosomal surface, and that TSC2 may negatively influence GTP-bound RHEB, these data suggest that resistance exercise may activate mTORC1 by alleviating its inhibition by TSC2. There is also evidence that the provision of amino acids promotes the translocation of mTOR to the lysosome, which may partially account for the synergistic impact of combining amino acid feeding and resistance exercise on MPS. Such a hypothesis awaits experimental confirmation in human models of exercise and feeding but presents an intriguing area of future research nonetheless.



6. TRANSLATIONAL RESPONSES TO RESISTANCE TRAINING

It is proposed that there are two ways in which resistance exercise enhances muscle hypertrophy. Firstly, enhance translational efficiency that is underpinned by an increase in the number of ribosomes binding to a single mRNA transcript, thus resulting in greater protein yield per mRNA. For example, ribosomes are typically spaced approximately 80–100 nucleotides apart, but following stimulation ribosomes can stack much closer, with reports of ribosomes capable of assembling only 27–29 nucleotides apart.⁷⁰ The second is enhanced translational capacity, which refers to the upregulation of ribosomal content. The ribosome consists of two subunits, the 40S and the 60S subunit, and is a ribonucleoprotein serving to increase protein content by the decoding of mRNA.^{71,72} Both of the subunits are comprised of a variety of ribosomal proteins and it is the increase in these ribosomal proteins and the ribosomal subunits in response to anabolic stimulation that is believed to act in concert with enhanced translational efficiency to promote increases in cell size. Ribosomal biogenesis is a highly controlled and complex biological process regulated in part via mTORC1.⁷¹ Indeed, distinct to many ribosomal protein and growth-related mRNAs is a unique terminal oligopyrimidine tract sequence (5'TOP) and these mRNAs are specifically regulated by mTORC1-P70S6K1 signaling.⁷³ Given that resistance exercise is a potent stimulator of both mTORC1 and MPS,⁷⁴ it has been proposed that resistance

exercise enhances rates of MPS in part via upregulation of ribosomal content.⁷¹

Whereas there is a significant body of research that has focused upon the role of ribosomal biogenesis in cardiac muscle, there is comparably less data available regarding the role of ribosomal biogenesis in skeletal muscle. A common readout of increased ribosomal biogenesis in skeletal muscle is an increase in total RNA content.⁵⁵ This assumption is based on the premise that the ribosome is ~30% protein and ~60% RNA and that the majority of RNA is ribosomal RNA (rRNA).^{71,75} In this regard, data from rodent models have shown elevations in rRNA content are associated with enhanced mTORC1 signaling.^{55,76} Additionally, treatment with rapamycin has been shown to not only prevent P70S6K1 phosphorylation but also inhibit load-induced increases in rRNA content.⁵⁵ Such findings would support the emerging thesis that, at least in models where constant load is applied, increases in load-induced MPS are by logical extension facilitated not only by enhanced translational efficiency but also by enhanced translational capacity.

We know that there is an incongruence between resistance exercise-induced increases in rates of MPS and the hypertrophic response to resistance exercise training in humans.⁷⁷ Thus, it is not known whether increases in translational capacity seen in rodents are also seen in humans. Potentially debunking this contention, Phillips *et al.*⁷⁸ reported that 20 weeks of supervised resistance exercise training actually downregulated the mRNA expression of the translational machinery. In this study,⁷⁸ it was shown that the individuals who demonstrated the greatest gains in skeletal muscle mass actually exhibited the biggest reductions in rRNA gene expression. Moreover, those that gained the greatest skeletal muscle mass also displayed a gene signature consistent with that of a rapamycin, in other words indicative of reduced mTORC1 activity. These data also would support other reports of incongruence between acute changes in kinase phosphorylation and skeletal muscle hypertrophy^{49,79} raising the question as to the relevance of assessing the phosphorylation/activity status of signaling molecules in the acute postexercise recovery period, at least in the context of skeletal muscle hypertrophy. However, it is important to acknowledge that despite rRNA expression negatively correlating with gains in muscle mass, the participants that responded the greatest to the resistance exercise-training program also tended to have higher levels of rRNA pretraining suggesting that these data are more indicative of enhanced translational efficiency. Thus, more work is clearly necessary in order to establish the role, if any, of increased ribosomal

abundance in long-term gains in human skeletal muscle mass in response to resistance exercise training in humans.



7. THE IMPACT OF AGING AND UNLOADING

Aging is associated with the progressive loss of skeletal muscle mass, termed sarcopenia.^{80,81} The limited capacity of older adults to mount a “youthful” protein synthetic response following ingestion of smaller doses of amino acids has already been well characterized and is proposed to be a main contributor to the sarcopenic condition.⁸² However, this process is confounded by the fact that older people fail to mount a similar protein synthetic response to a bout of resistance exercise,⁸³ resulting in compromised strength and muscle size gains when compared to their younger counterparts.⁸⁴ The “anabolic resistance” of aging skeletal muscle to an acute bout of resistance exercise in older adults has been shown to occur at a range of exercise intensities and is associated with a reduction in mTORC1-P70S6K1-related signaling.⁸⁵ Indeed, in older adults, the MPS response with performance of increasing intensities is sigmoidal, with minimal elevations occurring from 20% to 40% 1 RM, followed by a significant increase at 60% 1 RM and then no further enhancement at 75% or 90% 1 RM.⁸³ This MPS response is also accompanied by lower P70S6K1 and 4EBP1 phosphorylation 1 h postexercise.⁸³ Expanding upon these findings others have shown that activation of signaling proteins are lower even as long as 6 and 24 h following exercise,⁸⁵ thus providing one mechanism of age-related “anabolic resistance.” However, the etiology of anabolic resistance is likely multifactorial and complex in nature, the questions of how and why aging results in a lack of sensitivity to the normally potent anabolic stimulus of resistance exercise remain unknown. Discerning the relative contributions of age and physical inactivity toward the development of sarcopenia is challenging as they often are highly associated. For example, simply reducing the number of daily steps performed is known to result in a lower postprandial MPS response to ingestion of a protein-containing meal.⁸⁶ It could, we would contend, be argued that both physical inactivity and aging share similar hallmarks of sarcopenia and perhaps even a similar molecular signature; however, it is likely that a molecular aging program likely exists separate from inactivity *per se*. No doubt, experimental work is now needed to pinpoint the exact molecular mechanisms underpinning anabolic resistance in older adults.

Whereas resistance exercise results in a state of skeletal muscle hypertrophy, periods of muscle disuse induce a state of muscle atrophy.^{87–90} The mechanisms underpinning disuse muscle atrophy remain a topic of intense debate with some arguing that elevated rates of MPB result in the gradual losses of muscle mass⁹¹ whereas others suggest that it is a reduction in rates of MPS,^{92,93} particularly in response to amino acid ingestion. There are good arguments for both sides of these proposed mechanisms. Indeed, MPB is mediated by numerous processes such as the calcium-dependent calpain system^{94,95} and the ubiquitin proteasome pathway.⁹⁶ Many readouts suggestive of increased ubiquitin proteasome activity, such as the mRNA expression of muscle atrophy F-box/atrogin-1 (MAFBX) and muscle-specific-RING-finger protein 1 (MURF1), have been shown to be increased during early periods of disuse.^{96,97} However, it is not possible for the ubiquitin proteasome pathway to directly target contractile muscle proteins and other mediators of MPB that are required to induce full degradation of intact myofibrils. Moreover, the majority of data to support the assertion that muscle disuse atrophy is mediated by accelerated rates of MPB emanate from rodent models⁹⁶ of disuse or human models of disease⁹⁸ that do not accurately mimic uncomplicated human disuse atrophy. On the other hand, rates of MPS during muscle unloading are reduced in both the postabsorptive⁵⁹ and postprandial state⁹⁰ with immobilization. Reductions in rates of MPS that have been calculated to account for a significant proportion, but all, of the decline in muscle mass observed with disuse.^{26,93}

Rather than an either/or approach (MPS vs. MPB) as to the primary mechanism responsible for disuse atrophy, there is mounting evidence to suggest that both an increase in rates of MPB and a reduction in rates of MPS act in concert to reduce the size of skeletal muscle during periods of unloading.⁸⁷ Indeed, one proposition is that subsequent to immobilization there is an initial rapid but transient increase in rates of MPB that leads to a more sustained and prolonged reduction in rates of MPS. Support for this contention comes from data showing that elevations in the early stage of disuse could be targeting the degradation of translational proteins such as eukaryotic initiation factor 3F.^{99,100} The reduction in the content of these proteins subsequently serves to limit the capacity of the translational machinery to mount a sufficient response to anabolic stimulation necessary to sustain positive protein balance. Although to our knowledge, no study has identified if a period of muscle disuse in humans results in the targeted degradation of these proteins and, as such, this hypothesis awaits experimental confirmation.



8. SUMMARY

To conclude, resistance exercise-induced skeletal muscle hypertrophy is a complex process involving many signal transduction pathways. Unfortunately, despite many years of research the biological signal that translates the mechanical tension of resistance exercise (i.e., load) to these signal transduction pathways that subsequently activates the translational machinery, remains elusive. Current developments in mass spectrometry and stable isotope technology as well as use of the novel “omic” techniques will no doubt advance our understanding in the coming years. To date, the majority of research in the field has been generated from rodent models of overload for example using synergist ablation. These rodent models have provided valuable data regarding the mechanisms that regulate the size of skeletal muscle; however, there are fundamental differences between rodents and humans that need to be appreciated when interpreting such data. If we are to significantly enhance our understanding of the factors that control the size of human muscle mass both in response to muscle loading and unloading, more work needs to be conducted in human models of resistance exercise using a range of techniques to complement what we already know.

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REFERENCES

1. Egan B, Zierath JR. Exercise metabolism and the molecular regulation of skeletal muscle adaptation. *Cell Metab.* 2013;17(2):162–184.
2. Phillips SM. Physiologic and molecular bases of muscle hypertrophy and atrophy: impact of resistance exercise on human skeletal muscle (protein and exercise dose effects). *Appl Physiol Nutr Metab.* 2009;34(3):403–410.
3. Claassen H, Gerber C, Hoppeler H, Luthi JM, Vock P. Muscle filament spacing and short-term heavy-resistance exercise in humans. *J Physiol.* 1989;409:491–495.
4. Kelley G. Mechanical overload and skeletal muscle fiber hyperplasia: a meta-analysis. *J Appl Physiol* (1985). 1996;81(4):1584–1588.
5. Tipton KD, Ferrando AA, Phillips SM, Doyle Jr D, Wolfe RR. Postexercise net protein synthesis in human muscle from orally administered amino acids. *Am J Physiol.* 1999;276(4 pt 1):E628–E634.

6. [Witard OC, Tieland M, Beelen M, Tipton KD, van Loon LJ, Koopman R. Resistance exercise increases postprandial muscle protein synthesis in humans. *Med Sci Sports Exerc.* 2009;41\(1\):144–154.](#)
7. [Burd NA, West DW, Moore DR, et al. Enhanced amino acid sensitivity of myofibrillar protein synthesis persists for up to 24 h after resistance exercise in young men. *J Nutr.* 2011;141\(4\):568–573.](#)
8. [De Lorme TL, Watkins AL. Technics of progressive resistance exercise. *Arch Phys Med Rehabil.* 1948;29\(5\):263–273.](#)
9. [Cermak NM, Res PT, de Groot LC, Saris WH, van Loon LJ. Protein supplementation augments the adaptive response of skeletal muscle to resistance-type exercise training: a meta-analysis. *Am J Clin Nutr.* 2012;96\(6\):1454–1464.](#)
10. [American College of Sports Medicine. American college of sports medicine position stand. Progression models in resistance training for healthy adults. *Med Sci Sports Exerc.* 2009;41\(3\):687–708.](#)
11. [Schoenfeld BJ, Wilson JM, Lowery RP, Krieger JW. Muscular adaptations in low- versus high-load resistance training: a meta-analysis. *Eur J Sport Sci.* 2014;1–10.](#)
12. [Mitchell CJ, Churchward-Venne TA, West DWD, et al. Resistance exercise load does not determine training-mediated hypertrophic gains in young men. *J Appl Physiol.* 2012;113\(1\):71–77.](#)
13. [Burd NA, Holwerda AM, Selby KC, et al. Resistance exercise volume affects myofibrillar protein synthesis and anabolic signalling molecule phosphorylation in young men. *J Physiol.* 2010;588\(pt 16\):3119–3130.](#)
14. [Hubal MJ, Gordish-Dressman H, Thompson PD, et al. Variability in muscle size and strength gain after unilateral resistance training. *Med Sci Sports Exerc.* 2005;37\(6\):964–972.](#)
15. [Bamman MM, Petrella JK, Kim JS, Mayhew DL, Cross JM. Cluster analysis tests the importance of myogenic gene expression during myofiber hypertrophy in humans. *J Appl Physiol \(1985\).* 2007;102\(6\):2232–2239.](#)
16. [Hartman JW, Tang JE, Wilkinson SB, et al. Consumption of fat-free fluid milk after resistance exercise promotes greater lean mass accretion than does consumption of soy or carbohydrate in young, novice, male weightlifters. *Am J Clin Nutr.* 2007;86\(2\):373–381.](#)
17. [Burd NA, Moore DR, Mitchell CJ, Phillips SM. Big claims for big weights but with little evidence. *Eur J Appl Physiol.* 2013;113\(1\):267–268.](#)
18. [Kraemer WJ, Ratamess NA. Hormonal responses and adaptations to resistance exercise and training. *Sports Med.* 2005;35\(4\):339–361.](#)
19. [Bhasin S, Storer TW, Berman N, et al. The effects of supraphysiologic doses of testosterone on muscle size and strength in normal men. *N Engl J Med.* 1996;335\(1\):1–7.](#)
20. [Rennie MJ. Claims for the anabolic effects of growth hormone: a case of the emperor's new clothes? *Br J Sports Med.* 2003;37\(2\):100–105.](#)
21. [Doessing S, Heinemeier KM, Holm L, et al. Growth hormone stimulates the collagen synthesis in human tendon and skeletal muscle without affecting myofibrillar protein synthesis. *J Physiol.* 2010;588\(pt 2\):341–351.](#)
22. [West DW, Kujbida GW, Moore DR, et al. Resistance exercise-induced increases in putative anabolic hormones do not enhance muscle protein synthesis or intracellular signalling in young men. *J Physiol.* 2009;587\(pt 21\):5239–5247.](#)
23. [Fedele MJ, Lang CH, Farrell PA. Immunization against IGF-I prevents increases in protein synthesis in diabetic rats after resistance exercise. *Am J Physiol Endocrinol Metab.* 2001;280\(6\):E877–E885.](#)
24. [Spangenburg EE, Le Roith D, Ward CW, Bodine SC. A functional insulin-like growth factor receptor is not necessary for load-induced skeletal muscle hypertrophy. *J Physiol.* 2008;586\(1\):283–291.](#)

25. [Hamilton DL, Philp A, MacKenzie MG, Baar K. A limited role for PI\(3,4,5\)P3 regulation in controlling skeletal muscle mass in response to resistance exercise. *PLoS One*. 2010;5\(7\):e11624.](#)
26. [Phillips SM, Glover EI, Rennie MJ. Alterations of protein turnover underlying disuse atrophy in human skeletal muscle. *J Appl Physiol* \(1985\). 2009;107\(3\):645–654.](#)
27. [Cunningham ML. A mouse is not a rat is not a human: species differences exist. *Toxicol Sci*. 2002;70\(2\):157–158.](#)
28. [Greenhaff PL, Karagounis LG, Peirce N, et al. Disassociation between the effects of amino acids and insulin on signaling, ubiquitin ligases, and protein turnover in human muscle. *Am J Physiol Endocrinol Metab*. 2008;295\(3\):E595–E604.](#)
29. [Trommelen J, Groen B, Hamer H, de Groot LC, van Loon LJ. Mechanisms in endocrinology: exogenous insulin does not increase muscle protein synthesis rate when administered systemically: a systematic review. *Eur J Endocrinol*. 2015;173\(1\):R25–R34.](#)
30. [Kvornring T, Andersen M, Brixen K, Madsen K. Suppression of endogenous testosterone production attenuates the response to strength training: a randomized, placebo-controlled, and blinded intervention study. *Am J Physiol Endocrinol Metab*. 2006;291\(6\):E1325–E1332.](#)
31. [Schroeder ET, Villanueva M, West DD, Phillips SM. Are acute post-resistance exercise increases in testosterone, growth hormone, and IGF-1 necessary to stimulate skeletal muscle anabolism and hypertrophy? *Med Sci Sports Exerc*. 2013;45\(11\):2044–2051.](#)
32. [West DWD, Burd NA, Churchward-Venne TA, et al. Sex-based comparisons of myofibrillar protein synthesis after resistance exercise in the fed state. *J Appl Physiol*. 2012;112\(11\):1805–1813.](#)
33. [West DW, Burd NA, Tang JE, et al. Elevations in ostensibly anabolic hormones with resistance exercise enhance neither training-induced muscle hypertrophy nor strength of the elbow flexors. *J Appl Physiol* \(1985\). 2010;108\(1\):60–67.](#)
34. [Foster KG, Fingar DC. Mammalian target of rapamycin \(mTOR\): conducting the cellular signaling symphony. *J Biol Chem*. 2010;285\(19\):14071–14077.](#)
35. [Laplanche M, Sabatini DM. mTOR signaling in growth control and disease. *Cell*. 2012;149\(2\):274–293.](#)
36. [Inoki K, Li Y, Xu T, Guan KL. Rheb GTPase is a direct target of TSC2 GAP activity and regulates mTOR signaling. *Genes Dev*. 2003;17\(15\):1829–1834.](#)
37. [Inoki K, Zhu T, Guan KL. TSC2 mediates cellular energy response to control cell growth and survival. *Cell*. 2003;115\(5\):577–590.](#)
38. [Gwinn DM, Shackelford DB, Egan DF, et al. AMPK phosphorylation of raptor mediates a metabolic checkpoint. *Mol Cell*. 2008;30\(2\):214–226.](#)
39. [Inoki K, Li Y, Zhu T, Wu J, Guan KL. TSC2 is phosphorylated and inhibited by Akt and suppresses mTOR signalling. *Nat Cell Biol*. 2002;4\(9\):648–657.](#)
40. [Vander Haar E, Lee SI, Bandhakavi S, Griffin TJ, Kim DH. Insulin signalling to mTOR mediated by the Akt/PKB substrate PRAS40. *Nat Cell Biol*. 2007;9\(3\):316–323.](#)
41. [Hershey JW. Translational control in mammalian cells. *Annu Rev Biochem*. 1991;60:717–755.](#)
42. [Richardson CJ, Broenstrup M, Fingar DC, et al. SKAR is a specific target of S6 kinase 1 in cell growth control. *Curr Biol*. 2004;14\(17\):1540–1549.](#)
43. [Wang X, Li W, Williams M, Terada N, Alessi DR, Proud CG. Regulation of elongation factor 2 kinase by p90\(RSK1\) and p70 S6 kinase. *EMBO J*. 2001;20\(16\):4370–4379.](#)
44. [Gingras AC, Gygi SP, Raught B, et al. Regulation of 4E-BP1 phosphorylation: a novel two-step mechanism. *Genes Dev*. 1999;13\(11\):1422–1437.](#)

45. Laplante M, Sabatini DM. Regulation of mTORC1 and its impact on gene expression at a glance. *J Cell Sci.* 2013;126(pt 8):1713–1719.
46. Gkogkas C, Sonenberg N, Costa-Mattioli M. Translational control mechanisms in long-lasting synaptic plasticity and memory. *J Biol Chem.* 2010;285(42):31913–31917.
47. Dowling RJ, Topisirovic I, Fonseca BD, Sonenberg N. Dissecting the role of mTOR: lessons from mTOR inhibitors. *Biochim Biophys Acta.* 2010;1804(3):433–439.
48. Laplante M, Sabatini DM. mTOR signaling at a glance. *J Cell Sci.* 2009;122(pt 20):3589–3594.
49. Areta JL, Burke LM, Ross ML, et al. Timing and distribution of protein ingestion during prolonged recovery from resistance exercise alters myofibrillar protein synthesis. *J Physiol.* 2013;591(pt 9):2319–2331.
50. Moore DR, Robinson MJ, Fry JL, et al. Ingested protein dose response of muscle and albumin protein synthesis after resistance exercise in young men. *Am J Clin Nutr.* 2009;89(1):161–168.
51. Churchward-Venne TA, Burd NA, Mitchell CJ, et al. Supplementation of a sub-optimal protein dose with leucine or essential amino acids: effects on myofibrillar protein synthesis at rest and following resistance exercise in men. *J Physiol.* 2012;590(pt 11):2751–2765.
52. Mitchell CJ, Churchward-Venne TA, West DW, et al. Resistance exercise load does not determine training-mediated hypertrophic gains in young men. *J Appl Physiol (1985).* 2012;113(1):71–77.
53. Baar K, Esser K. Phosphorylation of p70(S6k) correlates with increased skeletal muscle mass following resistance exercise. *Am J Physiol.* 1999;276(1 pt 1):C120–C127.
54. Bodine SC, Stitt TN, Gonzalez M, et al. Akt/mTOR pathway is a crucial regulator of skeletal muscle hypertrophy and can prevent muscle atrophy in vivo. *Nat Cell Biol.* 2001;3(11):1014–1019.
55. Goodman CA, Frey JW, Mabrey DM, et al. The role of skeletal muscle mTOR in the regulation of mechanical load-induced growth. *J Physiol.* 2011;589(pt 22):5485–5501.
56. Drummond MJ, Fry CS, Glynn EL, et al. Rapamycin administration in humans blocks the contraction-induced increase in skeletal muscle protein synthesis. *J Physiol Lond.* 2009;587(7):1535–1546.
57. Gundermann DM, Walker DK, Reidy PT, et al. Activation of mTORC1 signaling and protein synthesis in human muscle following blood flow restriction exercise is inhibited by rapamycin. *Am J Physiol Endocrinol Metab.* 2014;306(10):E1198–E1204.
58. Choe AY, Yoon SO, Kim SG, Roux PP, Blenis J. Rapamycin differentially inhibits S6Ks and 4E-BP1 to mediate cell-type-specific repression of mRNA translation. *Proc Natl Acad Sci USA.* 2008;105(45):17414–17419.
59. Glover EI, Phillips SM, Oates BR, et al. Immobilization induces anabolic resistance in human myofibrillar protein synthesis with low and high dose amino acid infusion. *J Physiol.* 2008;586(pt 24):6049–6061.
60. Atherton PJ, Szewczyk NJ, Selby A, et al. Cyclic stretch reduces myofibrillar protein synthesis despite increases in FAK and anabolic signalling in L6 cells. *J Physiol.* 2009;587(pt 14):3719–3727.
61. Gordon SE, Fluck M, Booth FW. Selected contribution: skeletal muscle focal adhesion kinase, paxillin, and serum response factor are loading dependent. *J Appl Physiol (1985).* 2001;90(3):1174–1183. discussion 1165.
62. Fluck M, Carson JA, Gordon SE, Ziemiecki A, Booth FW. Focal adhesion proteins FAK and paxillin increase in hypertrophied skeletal muscle. *Am J Physiol.* 1999;277(1 pt 1):C152–C162.
63. de Boer MD, Selby A, Atherton P, et al. The temporal responses of protein synthesis, gene expression and cell signalling in human quadriceps muscle and patellar tendon to disuse. *J Physiol.* 2007;585(pt 1):241–251.

64. Crossland H, Kazi AA, Lang CH, et al. Focal adhesion kinase is required for IGF-I-mediated growth of skeletal muscle cells via a TSC2/mTOR/S6K1-associated pathway. *Am J Physiol Endocrinol Metab.* 2013;305(2):E183–E193.
65. Hornberger TA, Chu WK, Mak YW, Hsiung JW, Huang SA, Chien S. The role of phospholipase D and phosphatidic acid in the mechanical activation of mTOR signaling in skeletal muscle. *Proc Natl Acad Sci USA.* 2006;103(12):4741–4746.
66. You JS, Lincoln HC, Kim CR, et al. The role of diacylglycerol kinase zeta and phosphatidic acid in the mechanical activation of mammalian target of rapamycin (mTOR) signaling and skeletal muscle hypertrophy. *J Biol Chem.* 2014;289(3):1551–1563.
67. McGlory C, White A, Treins C, et al. Application of the [γ -³²P] ATP kinase assay to study anabolic signaling in human skeletal muscle. *J Appl Physiol.* 2014;116(5):504–513.
68. Jacobs BL, Goodman CA, Hornberger TA. The mechanical activation of mTOR signaling: an emerging role for late endosome/lysosomal targeting. *J Muscle Res Cell Motil.* 2014;35(1):11–21.
69. Jacobs BL, Goodman CA, Hornberger TA. The mechanical activation of mTOR signaling: an emerging role for late endosome/lysosomal targeting. *J Muscle Res Cell Motil.* 2013;35(1):11–21.
70. Wolin SL, Walter P. Ribosome pausing and stacking during translation of a eukaryotic mRNA. *EMBO J.* 1988;7(11):3559–3569.
71. Chaillou T, Kirby TJ, McCarthy JJ. Ribosome biogenesis: emerging evidence for a central role in the regulation of skeletal muscle mass. *J Cell Physiol.* 2014;229(11):1584–1594.
72. Mahoney SJ, Dempsey JM, Blenis J. Cell signaling in protein synthesis ribosome biogenesis and translation initiation and elongation. *Prog Mol Biol Transl Sci.* 2009;90:53–107.
73. Merrick WC. Eukaryotic protein synthesis: still a mystery. *J Biol Chem.* 2010;285(28):21197–21201.
74. Drummond MJ, Fry CS, Glynn EL, et al. Rapamycin administration in humans blocks the contraction-induced increase in skeletal muscle protein synthesis. *J Physiol.* 2009;587(pt 7):1535–1546.
75. Moss T, Langlois F, Gagnon-Kugler T, Stefanovsky V. A housekeeper with power of attorney: the rRNA genes in ribosome biogenesis. *Cell Mol Life Sci.* 2007;64(1):29–49.
76. Miyazaki M, McCarthy JJ, Fedele MJ, Esser KA. Early activation of mTORC1 signaling in response to mechanical overload is independent of phosphoinositide 3-kinase/Akt signalling. *J Physiol.* 2011;589(pt 7):1831–1846.
77. Mitchell CJ, Churchward-Venne TA, Parise G, et al. Acute post-exercise myofibrillar protein synthesis is not correlated with resistance training-induced muscle hypertrophy in young men. *PLoS One.* 2014;9(2):e89431.
78. Phillips BE, Williams JP, Gustafsson T, et al. Molecular networks of human muscle adaptation to exercise and age. *PLoS Genet.* 2013;9(3):e1003389.
79. Mitchell CJ, Churchward-Venne TA, Bellamy L, Parise G, Baker SK, Phillips SM. Muscular and systemic correlates of resistance training-induced muscle hypertrophy. *PLoS One.* 2013;8(10):e78636.
80. Janssen I. Evolution of sarcopenia research. *Appl Physiol Nutr Metab.* 2010;35(5):707–712.
81. Janssen I. The epidemiology of sarcopenia. *Clin Geriatr Med.* 2011;27(3):355–363.
82. Moore DR, Churchward-Venne TA, Witard O, et al. Protein ingestion to stimulate myofibrillar protein synthesis requires greater relative protein intakes in healthy older versus younger men. *J Gerontol A Biol Sci Med Sci.* 2014;70(1):57–62.

83. [Kumar V, Selby A, Rankin D, et al. Age-related differences in the dose-response relationship of muscle protein synthesis to resistance exercise in young and old men. *J Physiol.* 2009;587\(pt 1\):211–217.](#)
84. [Drummond MJ, Marcus RL, Lastayo PC. Targeting anabolic impairment in response to resistance exercise in older adults with mobility impairments: potential mechanisms and rehabilitation approaches. *J Aging Res.* 2012;2012:486930.](#)
85. [Fry CS, Drummond MJ, Glynn EL, et al. Aging impairs contraction-induced human skeletal muscle mTORC1 signaling and protein synthesis. *Skelet Muscle.* 2011;1\(1\):11.](#)
86. [Breen L, Stokes KA, Churchward-Venne TA, et al. Two weeks of reduced activity decreases leg lean mass and induces “anabolic resistance” of myofibrillar protein synthesis in healthy elderly. *J Clin Endocrinol Metab.* 2013;98\(6\):2604–2612.](#)
87. [Wall BT, Dirks ML, van Loon LJ. Skeletal muscle atrophy during short-term disuse: implications for age-related sarcopenia. *Ageing Res Rev.* 2013;12\(4\):898–906.](#)
88. [Suetta C, Hvid LG, Justesen L, et al. Effects of aging on human skeletal muscle after immobilization and retraining. *J Appl Physiol \(1985\).* 2009;107\(4\):1172–1180.](#)
89. [Dirks ML, Wall BT, Nilwik R, Weerts DH, Verdijk LB, van Loon LJ. Skeletal muscle disuse atrophy is not attenuated by dietary protein supplementation in healthy older men. *J Nutr.* 2014;144\(8\):1196–1203.](#)
90. [Wall BT, Snijders T, Senden JM, et al. Disuse impairs the muscle protein synthetic response to protein ingestion in healthy men. *J Clin Endocrinol Metab.* 2013;98\(12\):4872–4881.](#)
91. [Reid MB, Judge AR, Bodine SC. CrossTalk opposing view: the dominant mechanism causing disuse muscle atrophy is proteolysis. *J Physiol.* 2014;592\(pt 24\):5345–5347.](#)
92. [Phillips SM, McGlory C. Rebuttal from Stuart M. Phillips and Chris McGlory. *J Physiol.* 2014;592\(pt 24\):5349.](#)
93. [Phillips SM, McGlory C. CrossTalk proposal: the dominant mechanism causing disuse muscle atrophy is decreased protein synthesis. *J Physiol.* 2014;592\(pt 24\):5341–5343.](#)
94. [Bartoli M, Richard I. Calpains in muscle wasting. *Int J Biochem Cell Biol.* 2005;37\(10\):2115–2133.](#)
95. [Salem M, Nath J, Rexroad CE, Killefer J, Yao J. Identification and molecular characterization of the rainbow trout calpains \(Capn1 and Capn2\): their expression in muscle wasting during starvation. *Comp Biochem Physiol B Biochem Mol Biol.* 2005;140\(1\):63–71.](#)
96. [Bodine SC, Baehr LM. Skeletal muscle atrophy and the E3 ubiquitin ligases MuRF1 and MAFbx/atrogen-1. *Am J Physiol Endocrinol Metab.* 2014;307\(6\):E469–E484.](#)
97. [Murton AJ, Constantin D, Greenhaff PL. The involvement of the ubiquitin proteasome system in human skeletal muscle remodelling and atrophy. *Biochim Biophys Acta.* 2008;1782\(12\):730–743.](#)
98. [Pasini E, Aquilani R, Dioguardi FS, D’Antona G, Gheorghiadu M, Taegtmeier H. Hypercatabolic syndrome: molecular basis and effects of nutritional supplements with amino acids. *Am J Cardiol.* 2008;101\(11A\):11E–15E.](#)
99. [Csibi A, Leibovitch MP, Cornille K, Tintignac LA, Leibovitch SA. MAFbx/atrogen-1 controls the activity of the initiation factor eIF3-f in skeletal muscle atrophy by targeting multiple C-terminal lysines. *J Biol Chem.* 2009;284\(7\):4413–4421.](#)
100. [Csibi A, Cornille K, Leibovitch MP, et al. The translation regulatory subunit eIF3f controls the kinase-dependent mTOR signaling required for muscle differentiation and hypertrophy in mouse. *PLoS One.* 2010;5\(2\):e8994.](#)